

9 The Rotated-Pentagon Profile

The Haim–Kislev–Ostrov counterexample is a Lagrangian product of two regular pentagons. We determine how its capacity changes when one pentagon rotates relative to the other. The volume stays constant, so this also determines which relative positions maximize the systolic ratio.

Let $P_5 \subset \mathbb{R}^2$ be a regular pentagon centred at the origin with circumradius one. Write $R(\theta)$ for counterclockwise rotation through angle θ , and put

$$K_\theta = P_5 \times_L R(\theta)P_5 \subset \mathbb{R}_q^2 \oplus \mathbb{R}_p^2.$$

The Lagrangian product here is the Cartesian product in the two coordinate planes, with symplectic form $\omega_0((q, p), (q', p')) = q \cdot p' - p \cdot q'$. Simultaneous rotation of both planes is symplectic, so the choice of initial orientation for P_5 does not affect the capacity. We use

$$\text{sys}(K) = \frac{c_{\text{EHZ}}(K)^2}{2 \text{vol}_4(K)}.$$

The sampled ratios in Figure 1 rise from the aligned position to a maximum at $\theta = \pi/10$, then fall symmetrically. They suggested a profile proportional to $\sec^2 \theta$ on $[0, \pi/10]$. We prove this formula by comparing the quadratic programs at different angles. Their feasible weights remain the same under rotation, and each fixed choice has a particularly simple angular dependence. The known capacity at the Haim–Kislev–Ostrov position will then bound every such choice throughout the interval.

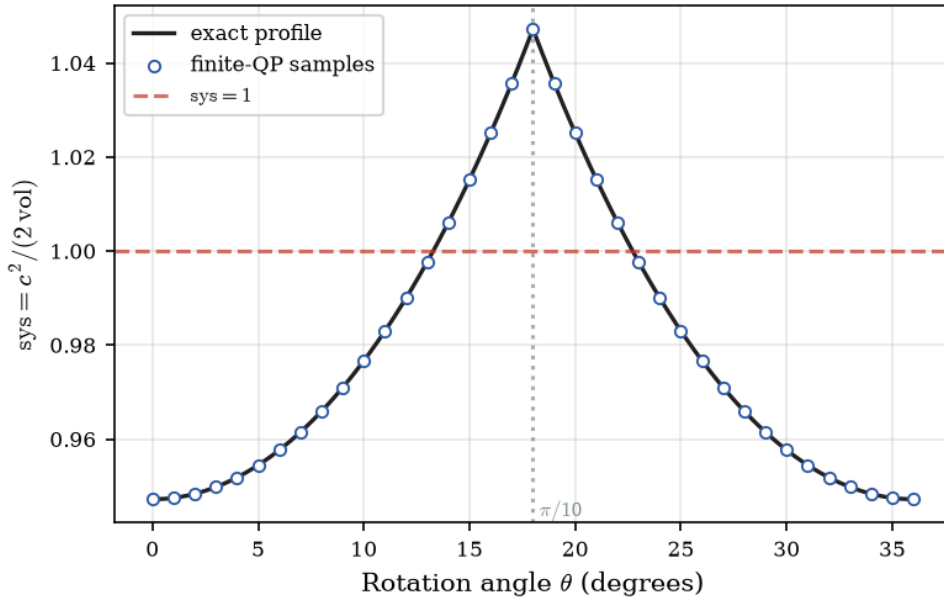


Figure 1: Systolic ratios for the rotating pentagon product. The blue markers are numerical evaluations of the finite quadratic programs; the black curve is the formula proved in this chapter. The horizontal line marks $\text{sys} = 1$.

To state the profile for all angles, define

$$d(\theta) = \min_{k \in \mathbb{Z}} \left| \theta - \frac{k\pi}{5} \right|.$$

This is the distance to the nearest multiple of $\pi/5$, and takes values in $[0, \pi/10]$.

Theorem 9.1 (Rotated-pentagon profile). *For every real θ ,*

$$c_{\text{EHZ}}(K_\theta) = (1 + \cos(\pi/5))^2 \sec d(\theta), \quad (1)$$

$$\text{sys}(K_\theta) = \frac{5 + 2\sqrt{5}}{10} \sec^2 d(\theta). \quad (2)$$

The maximum systolic ratio is $(3 + \sqrt{5})/5 > 1$, attained exactly at $\theta = \pi/10 + k\pi/5$, $k \in \mathbb{Z}$.

9.1 Symmetries and the Haim–Kislev–Ostrover position

We first reduce the angles to $[-\pi/10, \pi/10]$ and identify the capacity at its endpoints. The pentagon's rotational symmetry gives $K_{\theta+2\pi/5} = K_\theta$. If S is reflection in a symmetry axis of P_5 , the simultaneous reflection $(q, p) \mapsto (Sq, Sp)$ is symplectic and sends K_θ to $K_{-\theta}$. Thus capacity is even as a function of θ .

Exchanging the two factors gives a further symmetry. The symplectic map $(q, p) \mapsto (p, -q)$ sends K_θ to $R(\theta)P_5 \times_L (-P_5)$. Rotate both factors through $-\theta$ and use $-P_5 = R(\pi/5)P_5$ to obtain $K_{\pi/5-\theta}$. Together with evenness, this proves that capacity has period $\pi/5$. Every angle can therefore be reduced to $[-\pi/10, \pi/10]$ without changing capacity; evenness will give the final expression in terms of $d(\theta)$.

Haim–Kislev and Ostrover computed the capacity of the product with relative angle $-\pi/2$. Their convention for the symplectic form is $\sum_i dp_i \wedge dq_i = -\omega_0$. The map $(q, p) \mapsto (q, -p)$ changes this convention to ours and sends their body to $K_{\pi/2}$. Since $\pi/2 = \pi/10 + 2\pi/5$, their result [HO26, Proposition 1.4], together with the reflection symmetry, gives

$$c_{\text{EHZ}}(K_{-\pi/10}) = c_{\text{EHZ}}(K_{\pi/10}) = 2 \cos(\pi/10)(1 + \cos(\pi/5)). \quad (3)$$

9.2 Feasible weights under rotation

The quadratic capacity formula lets us compare different angles using the same facet order and weights. We spell out its variables for this family. Set

$$h = \cos(\pi/5), \quad n_i = \left(\cos\left(\frac{\pi}{2} + \frac{2\pi i}{5}\right), \sin\left(\frac{\pi}{2} + \frac{2\pi i}{5}\right) \right), \quad 0 \leq i < 5.$$

Choose the orientation $P_5 = \{q : n_i \cdot q \leq h, 0 \leq i < 5\}$. The ten facets of K_θ then have normalized normals

$$a_i(\theta) = h^{-1}(n_i, 0), \quad a_{i+5}(\theta) = h^{-1}(0, R(\theta)n_i), \quad 0 \leq i < 5,$$

so that their inequalities are $a_i(\theta) \cdot x \leq 1$.

A facet word $\sigma = (\sigma_1, \dots, \sigma_m)$ is an ordered list of distinct indices from $\{0, \dots, 9\}$. Its feasible weights satisfy

$$\beta_k \geq 0, \quad \sum_{k=1}^m \beta_k = 1, \quad \sum_{k=1}^m \beta_k a_{\sigma_k}(\theta) = 0. \quad (4)$$

The last equation is closure. For these weights, define

$$Q_{\sigma, \beta}(\theta) = \sum_{1 \leq j < k \leq m} \beta_j \beta_k \omega_0(a_{\sigma_j}(\theta), a_{\sigma_k}(\theta)). \quad (5)$$

This uses the traversal order of the quadratic-program chapter. The Haim–Kislev formula [Hai19] states that

$$c_{\text{EHZ}}(K_\theta) = \frac{1}{2Q_{\max}(\theta)}, \quad Q_{\max}(\theta) = \max_{\sigma, \beta} Q_{\sigma, \beta}(\theta) > 0, \quad (6)$$

where the maximum is over all words and their feasible weights. There are finitely many words, and each feasible set is a closed subset of a simplex, so the maximum exists. Zero weights allow facets to be omitted.

For this product, closure separates into equations in the two planes:

$$\sum_{\sigma_k < 5} \beta_k n_{\sigma_k} = 0, \quad R(\theta) \sum_{\sigma_k \geq 5} \beta_k n_{\sigma_k - 5} = 0.$$

Removing the invertible rotation from the second equation shows that feasibility is independent of θ . Thus a word and weights chosen at one angle remain feasible at every other angle.

We first exhibit a choice that gives the proposed capacity as an upper bound. Take

$$\sigma = (3, 8, 1, 0, 5, 6), \quad \beta = \frac{1}{4(1+h)}(2h, 2h, 1, 1, 1, 1).$$

These weights are positive and sum to one. The normal n_3 points opposite the bisector of n_0 and n_1 , and hence

$$n_0 + n_1 = -2hn_3.$$

This identity gives closure in both factors. To evaluate Q , group the consecutive entries (1, 0) and (5, 6) of the word. Normals from the same factor have zero symplectic pairing, so bilinearity lets us replace each group by its weighted sum. Writing

$$u = \frac{n_3}{2(1+h)}, \quad v = R(\theta)u,$$

the four weighted sums, in their order, are

$$(u, 0), \quad (0, v), \quad (-u, 0), \quad (0, -v).$$

Their ordered symplectic pairings sum to $2u \cdot v$. Therefore

$$Q_{\sigma, \beta}(\theta) = \frac{\cos \theta}{2(1+h)^2}. \tag{7}$$

For $|\theta| \leq \pi/10$ this is positive, and the capacity formula yields

$$c_{\text{EHZ}}(K_\theta) \leq (1+h)^2 \sec \theta. \tag{8}$$

To prove equality, we must show that no other feasible choice gives a larger quadratic value.

9.3 Bounding all choices by their endpoint values

Fix any facet word σ and feasible weights β . Same-factor pairings in (5) vanish, and each mixed pairing is, up to sign, $h^{-2}n_i \cdot R(\theta)n_j$. There are consequently constants $A_{\sigma, \beta}$ and $B_{\sigma, \beta}$ such that

$$Q_{\sigma, \beta}(\theta) = A_{\sigma, \beta} \cos \theta + B_{\sigma, \beta} \sin \theta.$$

This angular dependence determines the value inside our interval from the two endpoint values. The sine addition formulas give

$$\begin{aligned} Q_{\sigma, \beta}(\theta) &= \frac{\sin(\pi/10 - \theta)}{\sin(\pi/5)} Q_{\sigma, \beta}(-\pi/10) \\ &\quad + \frac{\sin(\pi/10 + \theta)}{\sin(\pi/5)} Q_{\sigma, \beta}(\pi/10). \end{aligned} \tag{9}$$

For $-\pi/10 \leq \theta \leq \pi/10$, both coefficients are nonnegative and their sum is $\cos \theta / \cos(\pi/10)$.

The fixed choice remains feasible at both endpoints, so its quadratic value there is at most the corresponding maximum. By (3) and (6),

$$Q_{\sigma,\beta}(\pm\pi/10) \leq Q_{\max}(\pm\pi/10) = \frac{1}{4\cos(\pi/10)(1+h)}.$$

We may substitute these upper bounds into (9) because its coefficients are nonnegative. This gives

$$Q_{\sigma,\beta}(\theta) \leq \frac{\cos \theta}{4\cos^2(\pi/10)(1+h)} = \frac{\cos \theta}{2(1+h)^2},$$

using $2\cos^2(\pi/10) = 1+h$. The bound holds for every feasible choice, and the choice in (7) attains it. We have proved

$$Q_{\max}(\theta) = \frac{\cos \theta}{2(1+h)^2}, \quad c_{\text{EHZ}}(K_\theta) = (1+h)^2 \sec \theta, \quad |\theta| \leq \pi/10.$$

Evenness and $\pi/5$ -periodicity give (1) for all real angles.

Finally, the five triangles joining the centre of P_5 to its edges give

$$\text{area}(P_5) = \frac{5}{2} \sin(2\pi/5), \quad \text{vol}_4(K_\theta) = \left(\frac{5}{2} \sin(2\pi/5) \right)^2.$$

Substitution into the definition of the systolic ratio yields

$$\text{sys}(K_\theta) = \frac{(1 + \cos(\pi/5))^4}{2 \left(\frac{5}{2} \sin(2\pi/5) \right)^2} \sec^2 d(\theta) = \frac{5 + 2\sqrt{5}}{10} \sec^2 d(\theta).$$

Since $\sec^2 t$ is strictly increasing on $[0, \pi/10]$, the maximum occurs precisely when $d(\theta) = \pi/10$. At those angles its value is $(3 + \sqrt{5})/5$. This completes the proof of Theorem 9.1.

References

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- [HO26] Pazit Haim-Kislev and Yaron Ostrover. “A counterexample to Viterbo’s conjecture”. In: *Annals of Mathematics* 203.2 (2026), pp. 603–622. DOI: 10.4007/annals.2026.203.2.5.